How much of a real fund depends on nature

The question

What share of a fund's market value sits in companies whose activity has a High or Very High ENCORE dependency rating on at least one ecosystem service, and on each of the 25 services? The fund is the State Street® SPDR® MSCI EMU UCITS ETF (ISIN IE00B910VR50), 213 holdings, EUR 348mn, as of 21-Oct-2026.

The answer

25.4% of market value. Interval 20.3% to 30.9% for the process mix inside each company's industry; 20.1% to 39.4% when the industry-to-ENCORE crosswalk is also uncertain. The cut-off decides more than either: Very High only gives 16.7%, Medium and above gives 68.7%. The 26% of the fund in financials scores 0%, because ENCORE rates a bank's own production process, not its borrowers.

What it is not

A direct, equity-side, process-level reading of ENCORE. It is not the ECB's loan-book figure (75% of euro area bank loans, with supply-chain links), and it is not a company-level measurement: no company reports its own dependency here.

Method: three joins and one draw

Inputs, all fetched by src/fetch.py

ENCORE knowledge base (JSON served to its public Explore tool): 3,386 ratings of 25 ecosystem services over 271 production processes. State Street holdings file for the fund, with GICS industry. ECB blog page (comparator only). SHA-256 of each file in results/fetch_log.json.

Join 1: holding to industry

213 holdings in 54 GICS industries. EUR 3.2mn of 'Unassigned' rows is dropped, not scored.

Join 2: industry to ENCORE

Each GICS industry maps to one or more ENCORE sub-industries (my labels, in GICS_TO_ENCORE). A company is treated as one of that set's production processes, equal probability. Variants: narrow (first sub-industry), base, broad (whole section).

Rating to flag

A process is dependent on a service if ENCORE rates it High or Very High (base).
'At least one service' is the OR over all services; per-service shares use one column.

Interval

5,000 draws. Each holding draws one process from its industry's mix, independently: a company's activity mix is its own. The crosswalk variant is drawn once per industry and shared by every holding in it: a wrong crosswalk is wrong for all of them.
The fund is not resampled; the estimand is this fund. Nothing here is forward-looking, so there is no backtest.

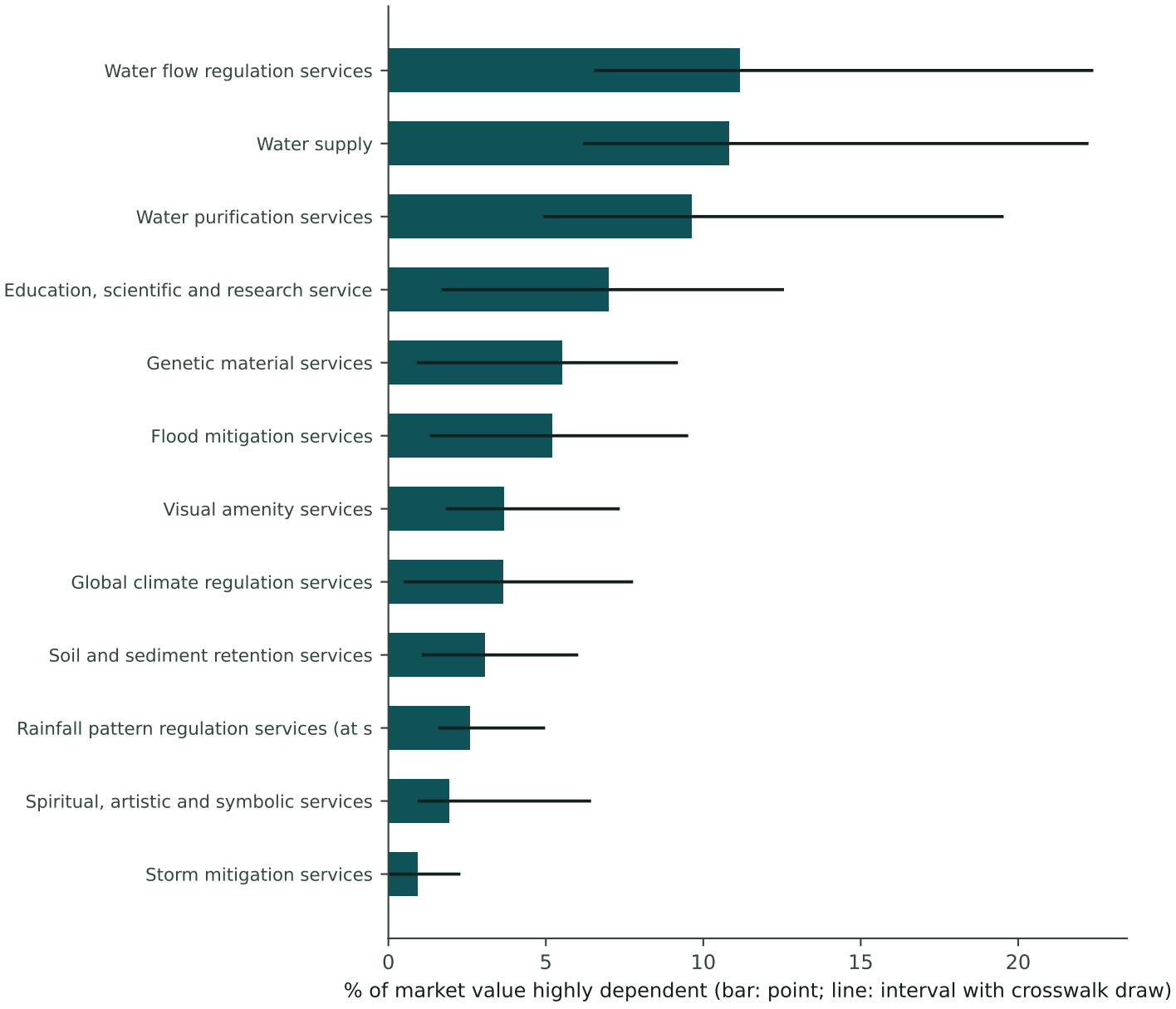
Result: water, and a few sectors, carry it

By service

Highest: Water flow regulation services 11.1% [6.6%, 22.3%],
Water supply 10.8%, Water purification services 9.6%.
20 of 25 services have any exposure at the High-and-above cut-off.
Services are not additive: the same company is dependent on several.

By sector, contribution to the headline

Utilities: 5.7% of the headline (weight 7%, own share 82%)
Health Care: 5.1% of the headline (weight 6%, own share 84%)
Consumer Staples: 3.5% of the headline (weight 6%, own share 62%)
Financials contributes 0.0% from 26% of the fund.



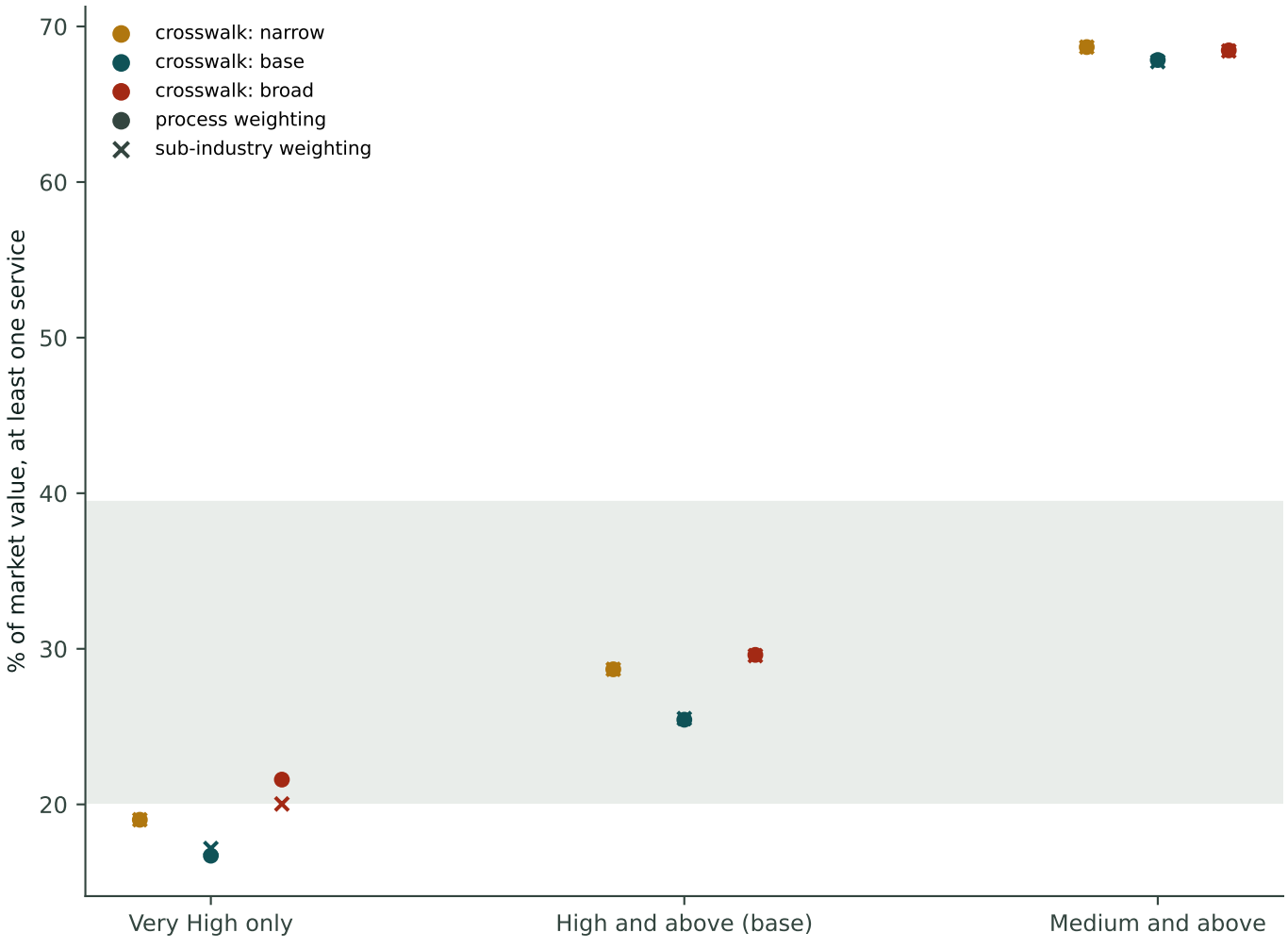
Sensitivity and validation: the cut-off is the finding

What moves the headline (18 combinations)

Rating cut-off: 16.7% to 67.8%, a range of 51.1 points.
Crosswalk variant: 25.4% to 29.6%, 4.2 points. Weighting rule: 0.07 points.
Across all 18, the headline runs 16.7% to 68.7%; the rank correlation of the 25 services against the base ordering is never below 0.74. Ordering is steadier than level.
Grey band: interval with crosswalk draw.

Validation

No independent measure of this fund's dependency exists. The ECB reports 75% of euro area bank loans and 72% of companies highly dependent on one service (score above 0.7, with supply chains, bank loans). My 25.4% is 49.6 points below that; even at Medium and above (68.7%) it is 6.3 points below. The estimands differ in portfolio, direct vs indirect links and score scale, so this is context, not a test of my number.



Limits, and what I would build next

The assumption that would break it

That a listed company's activity is a draw from its GICS industry's ENCORE processes and that ENCORE's direct rating is the dependency. Financials are 26% of the fund and score 0%; if they inherited the non-financial share (34.6%) the headline would rise by 9.2 points. The headline is a direct-dependency floor, not a risk estimate.

What a sceptic attacks first

The cut-off: 51 points between the strictest and loosest reading, from one rating scale that ENCORE assigns by expert judgement with no error attached. The interval does not include that error; the sweep is the only treatment of it.
Also: market-cap weights for a Eurozone large-cap fund say nothing about SMEs or loans.

Next

Company-level activity splits (revenue by NACE) in place of the uniform draw; supply-chain dependency through an input-output table; a loan-book view for a bank with a published sector breakdown.